QUICK EXAM REFERENCE

DBMS Normalization Guide

A complete, exam-oriented summary of 1NF through 5NF rules, dependencies, and memory tricks

1. First Normal Form (1NF)

Condition: Every attribute must contain **atomic (single) values**. No multi-valued or repeating groups inside a single cell.

❌ **Not in 1NF (Multi-valued cell)**

Roll_No	Name	Course
1	Ravi	**C, C++**
2	Aman	**Java, DBMS**

✅ **In 1NF (Atomic values)**

Roll_No	Name	Course
1	Ravi	C
1	Ravi	C++
2	Aman	Java
2	Aman	DBMS

🧠 **Golden Rule: 1NF = Atomicity (One Cell → One Value)**

2. Second Normal Form (2NF)

Conditions: Must be in **1NF** AND have **no partial dependency** of a non-prime attribute on a candidate key.

What is Partial Dependency?

Suppose AB is a Candidate Key ($AB \to C$). If part of the key determines a non-prime attribute ($A \to C$ or $B \to C$), it is a partial dependency (❌ Not 2NF).

⭐ **Golden Rule: 2NF = 1NF + No Partial Dependency**

⚡ **Important Shortcut:** If all candidate keys consist of a **single attribute**, partial dependency is impossible. Therefore: *1NF + Single-attribute candidate key ⇒ Automatically 2NF*.

3. Third Normal Form (3NF)

Conditions: Must be in **2NF** AND satisfy the 3NF FD condition.

For every non-trivial FD $X \to A$, either **X is a Superkey** OR **A is a Prime Attribute**.

Transitive Dependency: Occurs when $X \to Y$ and $Y \to Z$ (where X is key, Y & Z are non-prime), leading to $X \to Z$.

Example: $Roll\_No \to Dept\_ID$ and $Dept\_ID \to Dept\_Name$ creates redundant department data.

> ⭐ **Golden Rule: 3NF = 2NF + No Problematic Transitive Dependency**

> ⚠️ **Exam Trap:** Don't rely solely on "No non-prime attribute can determine anything." Always use the formal rule: **LHS is Superkey OR RHS is Prime**.

4. Boyce-Codd Normal Form (BCNF)

BCNF is a stricter version of 3NF. For every non-trivial FD $X \to Y$, **X MUST be a superkey** (no "RHS is prime" allowance).

3NF	BCNF
LHS is Superkey **OR** RHS is Prime	LHS **MUST** be a Superkey
Less strict; allows some non-superkey LHS if RHS is prime	More strict; completely forbids non-superkey LHS
Guarantees Lossless + Dependency Preservation	Guarantees Lossless, but may ❌ NOT preserve dependencies

> 🧠 **Golden Rule: BCNF = Every Determinant MUST be a Superkey**

5. Fourth Normal Form (4NF)

Condition: Must be in **BCNF** and have **no problematic non-trivial Multivalued Dependencies (MVDs)** ($X \twoheadrightarrow Y$).

Example: A person with 3 phone numbers and 3 emails stored in one table requires 9 Cartesian-product rows. Phone and Email are independent multi-valued facts.

Decomposition Solution: Split into two tables: $PERSON\_PHONE(Person, Phone)$ and $PERSON\_EMAIL(Person, Email)$.

> ⭐ **Golden Rule: 4NF = BCNF + No Problematic Multivalued Dependency**

6. Fifth Normal Form (5NF / PJ/NF)

Project-Join Normal Form: Deals with **Join Dependencies (JD).** A relation is in 5NF if it cannot be decomposed into smaller relations without losing data or introducing spurious tuples upon joining.

✅ **Lossless Decomposition**	❌ **Lossy Decomposition**
Original → Decompose → Join → **Exactly Original**	Original → Decompose → Join → **Original + Spurious Tuples**

⭐ **Golden Rule: 5NF = 4NF + No Problematic Join Dependency**

🔥 Complete Normal Form Hierarchy & Memory Tricks

Normal Form	Main Condition / Requirement	Removes / Controls	Memory Code
1NF	Atomic values in cells	Multi-valued cells / repeating groups	**1A** (Atomic)
2NF	1NF + No partial dependency	Partial dependency	**2P** (Partial)
3NF	2NF + (X superkey OR A prime)	Transitive dependency	**3T** (Transitive)
BCNF	3NF + Every determinant (X) is a superkey	Non-superkey determinants	**B-S** (Superkey)
4NF	BCNF + No problematic MVD ($X \twoheadrightarrow Y$)	Multivalued dependency	**4M** (Multi-valued)
5NF	4NF + No problematic Join Dependency	Join dependency	**5J** (Join)

SUPER-SHORT MEMORY TRICK & CHAIN

1NF: Atomic → **2NF: Partial** → **3NF: Transitive** → **BCNF: Superkey** → **4NF: MVD** → **5NF: Join**

Mnemonic: 1A → 2P → 3T → B-S → 4M → 5J

⚡ Final Exam-Ready Summary Answer

1. **1NF:** Atomic values only; no repeating groups or multi-valued cells.
2. **2NF:** In 1NF and contains no partial dependency of a non-prime attribute on any candidate key.
3. **3NF:** In 2NF and for every non-trivial FD $X \to A$, either X is a superkey or A is a prime attribute.
4. **BCNF:** In 3NF and for every non-trivial FD $X \to Y$, X must be a superkey.
5. **4NF:** In BCNF and contains no problematic non-trivial multivalued dependency.
6. **5NF:** In 4NF and contains no problematic join dependency (Project-Join Normal Form).